

Tenet: Agent Memory as a Self-Consistent World Model

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<https://github.com/Nas01010101/tenet>

Abstract

Long-term memory for LLM agents is almost universally implemented as *retrieval over a growing log of past turns*—a document-retrieval abstraction. We argue this is the wrong abstraction for an agent that must model a changing world. We introduce **knowledge churn**—the repeated updating of a fact over a long interaction—and show that retrieval-augmented memory (RAG-memory) *silently degrades* under it: as the number of stale versions of a fact exceeds the retrieval budget k , the reader is handed conflicting values and answers incorrectly. On a controlled benchmark, a strong RAG-memory falls from 100% to 50% current-value accuracy as a fact is updated $2 \rightarrow 12$ times. We propose **Tenet**, which reframes memory as a **self-consistent belief state**—a compact *world model of the user*—rather than a document store. Tenet (i) distills raw turns into atomic, keyed facts; (ii) maintains a **bi-temporal** record so a changed fact *supersedes* its predecessor (retired to history, not overwritten); (iii) enforces **belief–evidence consistency** by retiring raw evidence that echoes a superseded belief; and (iv) applies a **predictive-coding write policy**—surprise-gating—that stores only observations the model cannot already predict. Tenet holds 100% **current-value accuracy across all churn levels**, matches strong RAG on retrieval recall (95%), and attains the **best answer-accuracy-per-token** of the systems we evaluate. We report where strong RAG still wins—one-shot retrieval and multi-hop temporal synthesis—and release all code.

1 Introduction

An agent that talks to a user for months does not need a transcript; it needs a *model of the user* that stays true as the user changes. Yet the dominant memory design—retrieval-augmented generation over stored turns [1, 2]—treats memory as a document index: embed every turn, retrieve the top- k most similar, let the reader sort it out. This works for one-shot recall of a *static* fact. It fails, quietly, when facts **change**.

We formalize this as **knowledge churn**. A user who moves cities several times generates many “I moved to X ” turns; all are similar to “where do I live?”, so the top- k fills with *stale* versions. Once the number of updates exceeds k , the latest value may not be retrieved at all, and even when it is, the reader must infer recency from contradictory statements. Accuracy collapses (§4.2).

The root problem is abstraction. A document store has no notion that “I live in Boston” and “I live in Seattle” are the *same fact* with a *changed value*. We argue memory should be a **belief state**: a compact set of *current* beliefs, each with a temporal extent, updated by observation, kept internally consistent, and queryable across time—the stance world-model and predictive-coding accounts take toward perception [5], brought to agent memory.

Contributions.

1. We identify and name **knowledge churn**, a failure mode of retrieval memory, and give a controlled benchmark exhibiting it (RAG: 100% $\rightarrow$ 50%; §4.2).
2. We present **Tenet**, a memory that is a self-consistent belief state: bi-temporal supersession, a **belief-evidence consistency** rule, and a **surprise-gated** (predictive-coding) write policy.
3. We evaluate on LongMemEval-S and controlled tests: Tenet is **churn-robust** (100%), on par with RAG on recall (95%), and **best on accuracy-per-token**. We are explicit about where strong RAG still wins.

2 Related Work

Retrieval memory. Mem0 [1] distills salient facts at write time over a vector store; it attaches only a *creation* timestamp and *removed* its graph variant after finding it $3\times$ slower / $2\times$ tokens for a thin gain—evidence we weigh in choosing a light substrate. LongMemEval [2] is the standard long-horizon benchmark; its V2 adds a *latency-aware* metric, a field shift toward accuracy *per cost* that our per-token results target. **Temporal knowledge graphs.** Zep/Graphiti [3] maintain a bi-temporal graph with automatic invalidation—the closest prior work—but pay heavy per-write extraction and require graph infrastructure; Tenet keeps the bi-temporal semantics without the graph. **OS-style / observational memory.** MemGPT/Letta [4] page memory between context and archival storage; observational schemes keep a stable summary. Both are largely append-oriented and do not treat supersession or forgetting as first-class. **What is missing.** No prior system combines bi-temporal supersession, *belief-evidence consistency*, *predictive-coding write-gating*, and principled forgetting in a light, graph-free store—nor reports the knowledge-churn regime.

3 Method

Tenet stores two layers over one bi-temporal table: a **belief layer** of distilled, keyed facts, and an **evidence layer** of raw turns. Reads never call an LLM.

Distillation. Each turn is distilled into atomic facts, each with a stable semantic key $\kappa = \text{subject:attribute}$ (e.g. `user:residence`), a salience $s \in [0, 1]$, and an event time. The key makes supersession reliable: embedding similarity cannot separate a *restated* fact from a *value-changed* one (we measure a value change at cosine 0.99, indistinguishable from a paraphrase), but a shared key can.

Bi-temporal supersession. Every memory carries event time (`valid_at`, `invalid_at`) and transaction time (`created_at`, `expired_at`). Storing a fact at key κ whose value differs from the current one *supersedes* it: the old fact’s `invalid_at`/`expired_at` are set; it leaves the current set but stays in history. Current recall filters `expired_at` = $\emptyset$; `recall(as_of=t)` reconstructs the belief set at any past t (time-travel).

Belief-evidence consistency (key rule). The evidence layer answers detail questions but is where stale values hide: “I moved to Boston” survives after the belief moves on. We retire from current recall any raw slice e close to a *superseded* belief:

$$\text{exclude } e \iff \max_{f \in \mathcal{F}_{\text{expired}}} \cos(e, f) \geq \tau_{\text{stale}}, \quad \tau_{\text{stale}} = 0.80.$$

This single rule turns supersession into end-to-end correctness: 55% $\rightarrow$ 100% (§4.3).

Predictive-coding write policy. A world model stores *prediction error*. On write, a raw observation e is discarded if the store already predicts it:

$$\text{store } e \iff \max_{e' \in \mathcal{E}_{\text{store}}} \cos(e, e') < g_{\text{surprise}}, \quad g_{\text{surprise}} = 0.97.$$

This bounds the store and drops redundant repetition with no accuracy loss (§4.3).

Forgetting. Rank = relevance $\times$ decay, with $d = 2^{-\Delta t/h} (1 + \beta \log(1 + \text{uses}))$ ($0.6 + 0.8s$), half-life $h = 14$ d; a sweep archives low- d unpinned memories. Retrieval is a *dual pool* (beliefs for consistency, evidence for verbatim detail), each guaranteed a budget share.

4 Experiments

Protocol. LongMemEval_S (500 questions, ~ 115 k-token histories). We use a **gpt-4o reader** (the same reader Mem0 and Zep report against), a local embedder (**bge-small**), and a cheap **gpt-4o-mini** distiller; the shipped system runs the same code on Qwen Cloud (**text-embedding-v4**, **qwen3.7-plus**) by a config flip. Numbers are compared only to baselines run under identical settings. Baselines: **RAG** (top- k raw turns) and **full-context** (entire history).

4.1 Retrieval recall and the accuracy-per-token frontier (n=40, gpt-4o reader)

System	recall@10	QA acc	reader tok	acc/1k tok
full-context	—	65%*	~ 124 k	0.5*
RAG	95%	65%	2,101	30.9
Tenet	97.5%	52.5%	1,067	49.2

Tenet matches RAG on recall and gives the best accuracy per token. On raw accuracy it is beaten by a strong RAG (52.5 vs 65); the gap is in *multi-session* (28.6 vs 57.1) and single-session assistant/preference, where compression drops detail though recall is 95–100%—the loss is compression, not retrieval. Tenet gives the best accuracy-per-token ($1.6\times$ RAG). *Full-context, at a weaker reader, spends $100\times$ the tokens for no gain over RAG. The pattern is reader-robust: with a frontier reader (**claude-opus-4.8**) it is unchanged—RAG 67.5 / Tenet 57.5, acc/1k-tok 32.1 / 53.9 ($1.7\times$).

4.2 Knowledge churn (headline)

One fact updated N times amid distractors, $k = 6, 12$ principals/point:

updates N	2	4	6	8	10	12
RAG	100	100	100	67	58	50
Tenet	100	100	100	100	100	100

RAG degrades monotonically once $N > k$ (-50 pp); Tenet is flat at 100%. Supersession keeps exactly one current value regardless of churn—the property a *belief state* has and a document index cannot. **The curves are identical under a gpt-4o-mini and a gpt-4o reader:** the failure is structural (once $N > k$ the latest value is not reliably retrieved), so a stronger reader cannot rescue RAG.

4.3 Ablations

Belief–evidence consistency. Removing the rule drops Tenet to 55% current-value accuracy with a 45% stale-leak; adding it restores 100% / 0% leak—the single most important mechanism. **Surprise-gating** discards 15% of observations as redundant with no accuracy change, yielding a bounded store where RAG grows unboundedly.

5 Limitations

Not a better one-shot retriever—a well-tuned RAG matches or beats Tenet on static recall; Tenet’s edge is churn-robustness, per-token efficiency, and capabilities RAG lacks. **Multi-hop temporal synthesis** is the weakest category: distillation compresses away fine detail even when the right session is retrieved; a promising fix is *query-aware evidence expansion*. **Evaluation:** $n=40$, off-Qwen (gpt-4o reader, local embedder); the shipped system uses Qwen Cloud. Temporal-reasoning, weak under a mini reader, recovers to 40% under gpt-4o.

6 Conclusion

Treating agent memory as a **self-consistent belief state** rather than a document index makes it robust to the way real knowledge behaves: it changes. Tenet stays correct under knowledge churn where retrieval memory collapses, at the best accuracy-per-token we tested, with a light graph-free substrate and no LLM in the read path—and gets time-travel and principled forgetting for free. We hope *knowledge churn* becomes a standard axis for evaluating agent memory.

References

- [1] Chhikara et al. *Mem0: Production-Ready AI Agents with Scalable Long-Term Memory*. arXiv:2504.19413, 2025.
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- [5] Friston. *The free-energy principle: a unified brain theory?* Nat. Rev. Neurosci., 2010.